

Experiment 30

Aim

To perform morphological dilation on an image using OpenCV.

Procedure

1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a morphological kernel.
5. Apply the Dilation operation using cv2.dilate().
6. Display the original and dilated images.
7. Save the resulting image.

Code:

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
dilated = cv2.dilate(gray, kernel, iterations=1)
cv2.imshow("Original Image", img)
cv2.imshow("Dilated Image", dilated)
cv2.imwrite("dilated_image.jpg", dilated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The morphological dilation operation was successfully performed using OpenCV. The dilation increased the size of bright/white regions and expanded the boundaries of objects in the image.